

RTLs for instructions

Clock	Micro operations
T0	$AR \leftarrow PC$
T1	$IR \leftarrow M[AR], PC \leftarrow PC + 1$
T2	$AR \leftarrow IR(0-11), \text{Decode } (12-14) - D0 \rightarrow D7$

LDA (D2)

Clock	Micro operations
T3	$DR \leftarrow M[AR]$
T4	$AC \leftarrow DR, SC \leftarrow 0$

XOR (D0)

Clock	Micro operations
T3	$DR \leftarrow M[AR]$
T4	$AC \leftarrow DR \text{ XOR } AC, SC \leftarrow 0$

ADD (D1)

Clock	Micro operations
T3	$DR \leftarrow M[AR]$
T4	$AR \leftarrow DR + AC, SC \leftarrow 0$

STA (D3)

Clock	Micro operations
T3	$M[AR] \leftarrow AC, SC \leftarrow 0$

DIV(D5)

Clock	Micro operations
T3	$DR \leftarrow M[AR]$
T4	$AC \leftarrow AC / DR, SC \leftarrow 0$

ISZ(D6)

Clock	Micro operations
T3	$DR \leftarrow M[AR]$
T4	$DR \leftarrow DR + 1$
T5	$M[AR] \leftarrow DR$, If $(DR == 0), PC \leftarrow PC + 1, SC \leftarrow 0$

BUN (D4)

Clock	Micro operations
T3	$PC \leftarrow AR, SC \leftarrow 0$

Register load,increment and clear

Register	Load	Increment	Clear
DR	$(D0 + D1 + D2 + D5 + D6)T3$	$D6T4$	---
AR	$T0 + T2$	---	---
IR	$T1$	---	---
AC	$(D0 + D1 + D2 + D5)T4$	---	---
PC	$D4T3$	$T1 + D6T5(DR_ZERO)$	---
SC	---	---	$D0T4 + D1T4 + D2T4 + D3T3 + D4T3 + D5T4 + D6T5$

Memory read and write

Memory	
Read	$T1 + (D0 + D1 + D2 + D5 + D6)T3$
Write	$D3T3 + D6T5$

Bus select line, Decoder:

Encoder	Micro operation	S2	S1	S0	Bus
x1	$T1 + (D0 + D1 + D2 + D5 + D6)T3$	0	0	1	Memory
x2	$D4T3$	0	1	0	AR
x3	$T0$	0	1	1	PC
x4	$D1T4 + D6T6$	1	0	0	DR
x5	$D3T3$	1	0	1	AC
x6	$T2$	1	1	0	IR

ALU select lines

Operation	Micro operation	C2	C1	C0
XOR	$D0T4$	1	1	1
ADD	$D1T4$	0	0	1
DIV	$D5T4$	1	0	1
PASS	$D2T4$	0	1	0

C2,C1,C0 SOP:

C0	$T4(D0+D1+D5)$
C1	$T4(D0+D2)$
C2	$T4(D0+D5)$

Memory:

It was stored from memory address zero till fifteen in hexadecimal:

**20c0 d0 10c0 30c0 20c0 50d0 10e0 30e0
60f0 4000 0 0 f 5 0 ffd**